22PH202 – ELECTROMAGNETISM AND MODERN PHYSICS

Unit-V: ENERGY BANDS IN SOLIDS_LM1

Band Theory of Solids: Classification of Materials

Introduction to Band Theory

The band theory of solids explains the behavior of electrons in solid materials. According to this theory, the energy levels of electrons in solids form bands, and the electrical properties of materials are determined by the position and occupancy of these bands.

Energy Bands in Solids

Valence Band: The energy band containing the valence electrons, which are the outermost electrons involved in chemical bonding.

Conduction Band: The energy band above the valence band where electrons can move freely, contributing to electrical conductivity.

Band Gap: The energy gap between the valence band and the conduction band. The size of the band gap is crucial in determining the electrical properties of the material.

Classification of Materials Based on Band Theory

Materials are classified into conductors, semiconductors, and insulators based on the band gap and the arrangement of electrons in the bands.

Conductors (Metals) :

Band Structure: In conductors, the valence band and conduction band overlap or the conduction band is partially filled.

Electrical Properties: Electrons can move freely within the conduction band, allowing for high electrical conductivity.

Examples: Copper, aluminum, gold.

Semiconductors:

Band Structure: Semiconductors have a small band gap (typically less than 3 eV) between the valence band and the conduction band.

Electrical Properties: At absolute zero, the conduction band is empty, and the valence band is full, making the material an insulator. However, at room temperature, some electrons gain enough energy to jump across the band gap to the conduction band, allowing for limited electrical conductivity.

Intrinsic Semiconductors: Pure semiconductors without any impurity atoms (e.g., silicon, germanium).

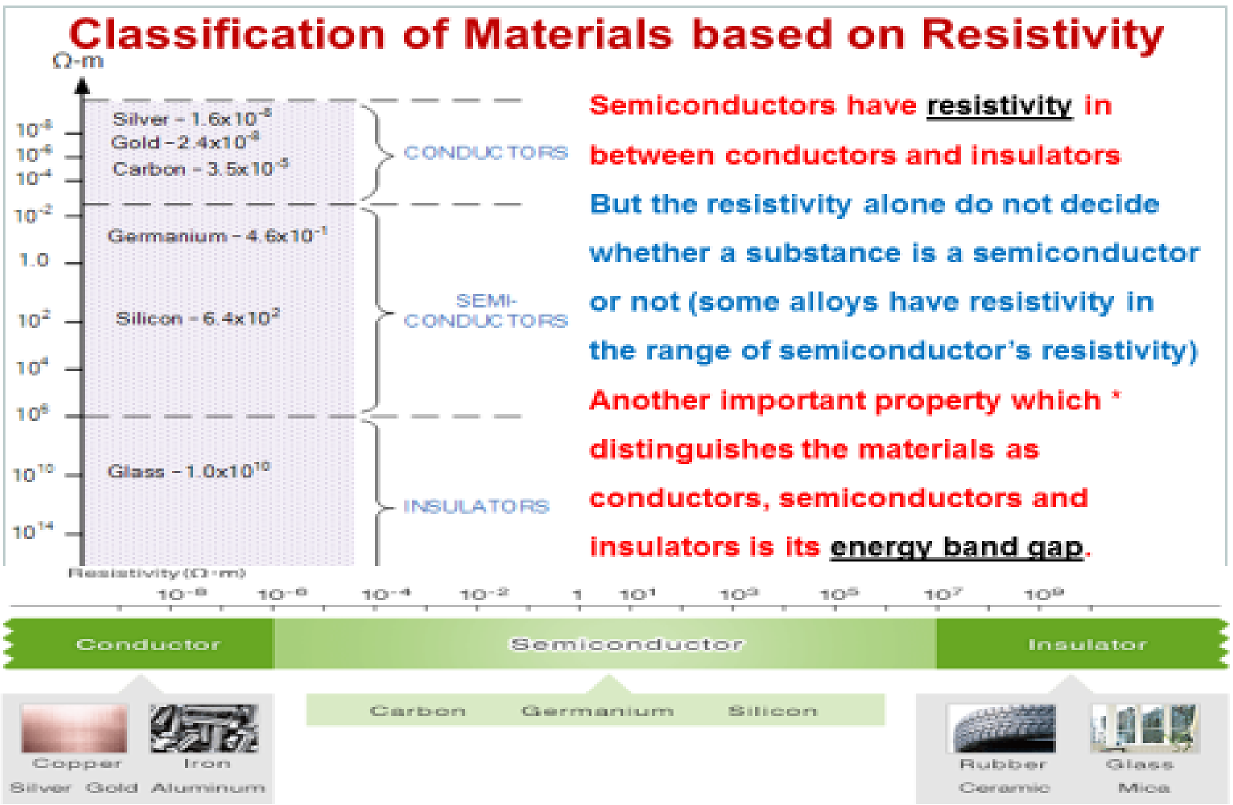
Extrinsic Semiconductors: Semiconductors doped with impurity atoms to enhance conductivity (e.g., n-type and p-type silicon).

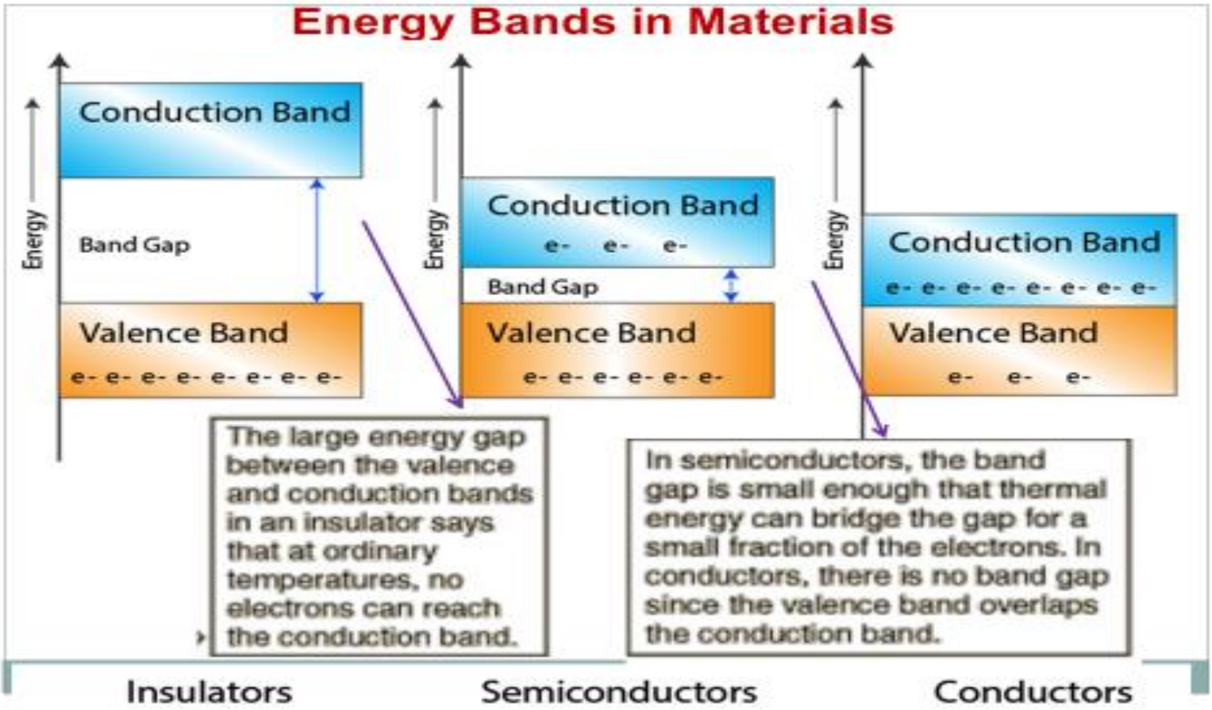
Insulators :

Band Structure: Insulators have a large band gap (typically greater than 3 eV).

Electrical Properties: The large band gap prevents electrons in the valence band from gaining enough energy to move to the conduction band, resulting in very low electrical conductivity.

Examples: Diamond, glass, rubber.





Semiconductor

Semiconducting materials having electrical conductivity between a good conductor and a good insulator. It is called semiconductor.

It is a special class of material which is very small in size and sensitive to heat.

The invention of semiconductors opened a new branch of technology called **solid state electronics.**

It leads to the development of ICs, microprocessors, computers and supercomputers.

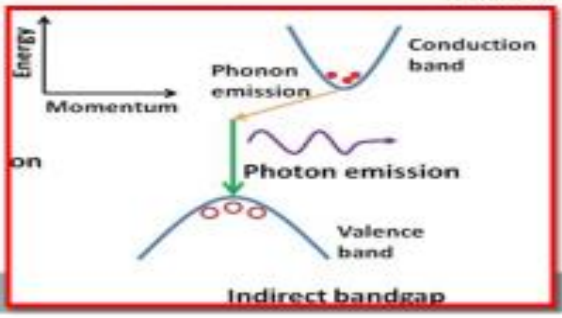
In short, semiconductor play a vital role in almost all advanced electronic devices.

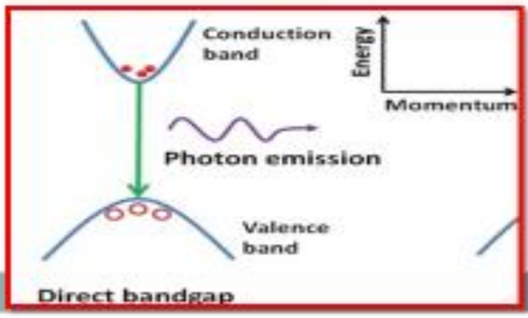
Properties of Semiconductor

1. The resistivity of semiconductors lies between conducting and insulating materials (10^{-4} to 0.5 ohm-m)
2. At T = 0 K, semiconductors behave as insulators
3. With increasing temperature or adding impurities, the conductivity of semiconductors increases (resistivity decreases) $\rho \propto \frac{1}{T}$
4. Semiconductors have negative temperature coefficient of resistance
5. Total conductivity $\sigma_{total} = \sigma_{electrons} + \sigma_{holes}$

Classification of Semiconductors based on Composition

1. Elemental Semiconductors, 2. Compound Semiconductors





Differentiate between elemental and compound semiconductors

S.No	Element Semiconductor	Compound Semiconductor
1	It is made by single element	They are formed by compound elements
2	Recombination of electron and hole takes pace indirectly.	Recombination of electron and hole takes pace directly with each other.
3	They are called as indirect bandgap semiconductor.	They are called as direct bandgap semiconductor.
4	Its energygap and carrier mobility are low	Its energygap and carrier mobility are high
5	During recombination, phonons are emitted	During recombination, photons are emitted
6	Life time of charge carries and current amplification are high.	Life time of charge carries and current amplification are low.
7	They are widely used in the production of diodes and transistors.	They are used for the manufacture of LEDs laser diodes etc.
8	Ex: Ge, Si etc	Ex: GaAs GaP

CO3: elemental and compound s